

Chat Session: Feature Engineering for ICU Mortality & Hemodynamic Instability

User: feature engineering practices for classifiers for ICU mortality or hemodynamic instability on clinical data (ABP, ECG, and PPG waveform from ICU recordings)? best python tools

Assistant: Feature engineering on high-frequency ICU physiological waveforms (ECG, ABP, PPG) bridges the gap between raw, noisy time-series data and clinical classifiers designed to predict mortality or acute hemodynamic instability.

1. Feature Engineering Practices

Features are typically engineered using a sliding window approach across specific observation intervals leading up to a specific prediction horizon.

Electrocardiogram (ECG)

- **Heart Rate Variability (HRV):** Extract time-domain indices (RMSSD, SDNN), frequency-domain components (LF, HF, LF/HF ratio), and non-linear properties.
- **Morphological Anomalies:** Track duration and amplitudes of P-wave, QRS complex, and T-wave (e.g., ST-segment shifts).
- **Ectopic Beat Burden:** Count premature ventricular contractions (PVCs) to assess electrical instability.

Arterial Blood Pressure (ABP)

- **Beat-by-Beat Continuous Vital Signs:** Track SBP, DBP, and calculate Mean Arterial Pressure (MAP).
- **Pulse Pressure (PP) & Shock Index (SI):** Compute PP (SBP - DBP) and Shock Index (HR / SBP) as an early warning for shock.
- **Waveform Structural Landmarks:** Measure slope of systolic upstroke, timing of dicrotic notch, and area under diastolic decay curve.

Photoplethysmography (PPG)

- **Pulse Rate Variability (PRV):** Serves as a surrogate measure derived from peripheral tissue perfusion.
- **Morphological Surrogates:** Locate systolic peak, dicrotic notch, and diastolic peak.
- **Pulse Oximetry Dynamics:** Extract baseline structural shifts (DC component) to quantify respiratory distress.

Cross-Waveform & Multimodal Features

- **Pulse Transit Time (PTT):** Compute the exact time lag between the R-peak of the ECG and the corresponding peak of PPG/ABP.

- **Signal Quality Indices (SQI):** Calculate window-level SQIs (skewness, kurtosis, power spectral density) to filter out artifacts.

2. Best Python Tools & Packages

- **NeuroKit2:** The gold standard toolbox for cleaning, peak detection, and extracting comprehensive HRV/PRV indices.
- **BIOBSS:** Open-source pipeline-centric package tailored precisely for multi-signal data and SQI assessments.
- **HeartPy:** Highly specialized library for parsing noisy, segment-level PPG or ECG data.
- **tsfresh:** Automatically extracts thousands of time-series features from windowed waveforms.
- **scikit-learn & XGBoost:** Highly favored frameworks for downstream classification on structured tabular features.
- **SHAP:** Crucial for deployment to explain black-box classifier predictions to clinicians transparently.

User: practices to predict which features to be most predictive

Assistant: Predicting which engineered features will be the most predictive before running your main classifier saves significant computational time. This is known as feature selection.

1. Filter Methods (Fast & Baseline)

- **Variance Thresholding:** Discard features that barely change across your patient cohort.
- **Mutual Information (MI):** Measures non-linear shared information between features and target variables.
- **Clinical Domain Filtering:** Prioritize features with established physiological backing (e.g., Shock Index > 0.7).

2. Wrapper Methods (Model-Guided)

- **Recursive Feature Elimination (RFE):** Recursively prunes the weakest features based on model feature rankings.
- **Sequential Feature Selection (SFS):** Sequentially adds or removes features based on optimal subset performance.

3. Embedded Methods & Regularization

- **Lasso Regression (L1 Regularization):** Drives coefficients of less predictive or highly correlated features exactly to zero.
- **Tree-Based Feature Importances:** Train LightGBM or XGBoost and check the built-in feature importances (specifically 'gain').

4. Advanced Clinical Practices

- **SHAP (SHapley Additive exPlanations):** Maps feature directionality and physiological impact.
- **Stability Selection:** Run selection algorithms across multiple bootstrapped subsamples to ensure features withstand noise.

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